Combined model/monitor interpolation of air quality design values using Empirical Bayesian Kriging to create background concentration grids for use in New Source Review air quality permit modeling

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STATE OF IDAHO
DEPARTMENT OF
ENVIRONMENTAL QUALITY

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Introduction

- 2019: Interagency group developed background concentrations for all criteria pollutants (data covered 2014-2017) using monitor data and AIRPACT5 model data
- Results posted here: Background Concentrations 2014 – 2017
  - Average usage 2000x/year, 5x/day
- 2024: Effort to update the BCs with more recent data began.
- Convened a BC Workgroup
- Sharing results for $PM_{2.5}$ and O_3

Objective

- The aim of the project is to produce updated grids of ambient background concentrations of criteria air quality pollutants for use in New Source Review (NSR) air quality permit modeling.
- Background concentrations are based on a definition in Appendix W of the Clean Air Act:
  - That portion of the background attributable to natural sources, other unidentified sources in the vicinity of the project, and regional transport contributions from more distant sources (domestic and international). The ambient contributions from these sources are typically accounted for through use of ambient monitoring data or, in some cases, regional-scale photochemical grid modeling results (40 CFR 51, Appendix W, section 8.3.1(a, ii)).

Methodology: Data Inputs

- Monitor data:
  - 110 $PM_{2.5}$ AQS monitors, 2 non-AQS sensors (WA)
  - 31 O_3 AQS monitors
  - 2022-2024 design values calculated for NAAQS form of $PM_{2.5}$ annual, $PM_{2.5}$ daily, and O_3 MDA8
  - $PM_{2.5}$ data excluded wildfire dates:
    - July 18 - October 23, 2022
    - May 17-20 and July 16 - September 23, 2023
    - July 4 – October 16, 2024.
  - O_3 data had no exclusions

Methodology: Data Inputs

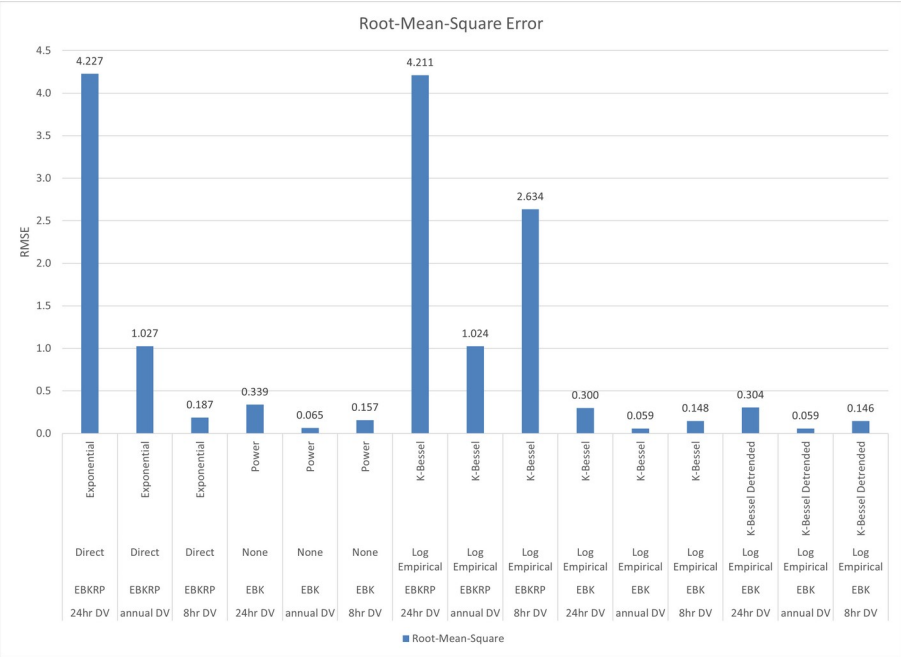
- Model data:
  - NOAA National Air Quality Forecast Capability (NAQFC)
  - Bias-corrected NOAA AQM v6 Day 1 at 5.079 km resolution
  - Design values calculated the same as monitor data
  - $PM_{2.5}$ DVs excluded same wildfire days as monitor data
  - O_3 data had no exclusions

Methodology: Interpolation

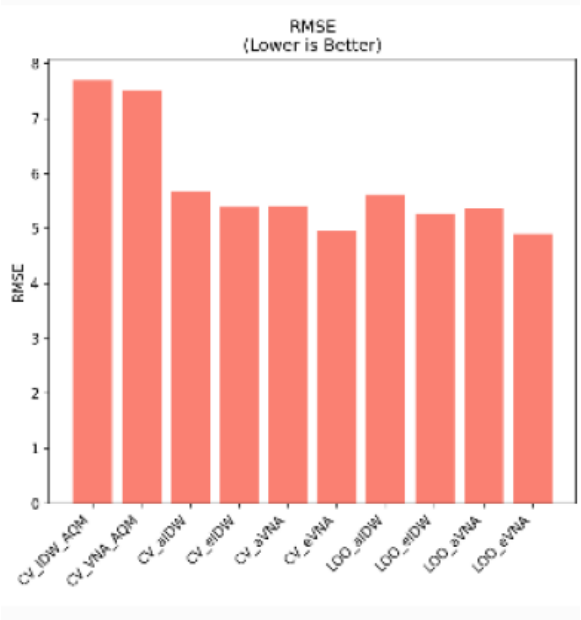
- Multiple interpolators were investigated:
  - Empirical Bayesian Kriging (EBK) and Empirical Bayesian Kriging with Regression Prediction (EBKRP), in GIS
  - Inverse Distance Weighting (IDW) and Voronoi Nearest Neighbor (VNA), in Python
  - Root Mean Square Error (RMSE) results were compared to choose the best interpolation model

Methodology: Interpolation

RMSE for EBK and EBKRP



RMSE for IDW and VNA



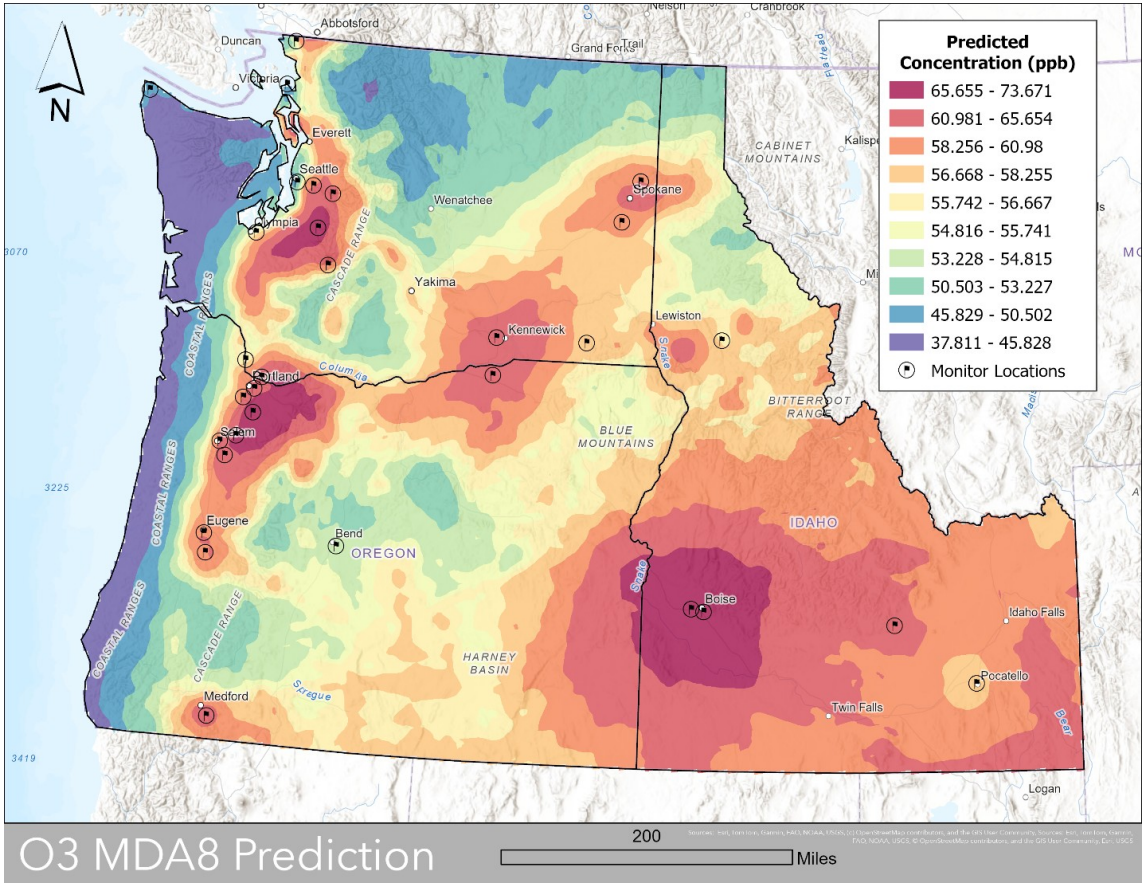
Methodology: EBK

- EBK was selected as the interpolator having the lowest RMSE
- Geostatistical method that analyzes the spatial pattern and distance between sampled sites and uses this information to build statistical models that represent spatial relationship between known values
- Subsets the input data and fits semivariogram models to each subset
- Determines the best model fit by running many iterations in a stochastic process
- For air quality pollutants like PM2.5, which have high local variability in concentration due to differing source distribution and interacting factors like meteorology and terrain, EBK's subsetting and local modeling is an advantage.

Results: Prediction Grids

O_3

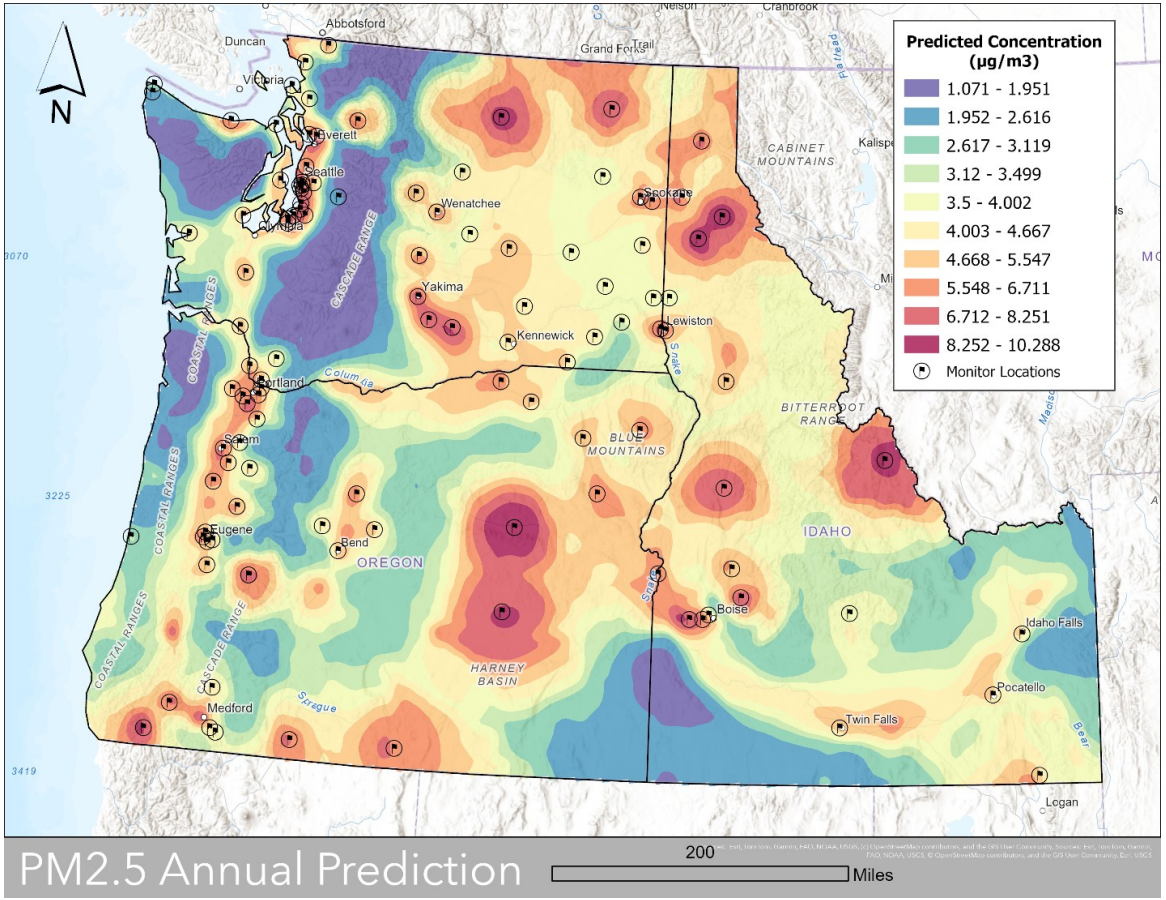
Output range	Monitor input	Model input
38-74 ppb	47-73 ppb	37-74 ppb



Results: Prediction Grids

$PM_{2.5}$ annual

Output range	Monitor input	Model input
1-10 µg/m³	1-12 µg/m³	1-10 µg/m³

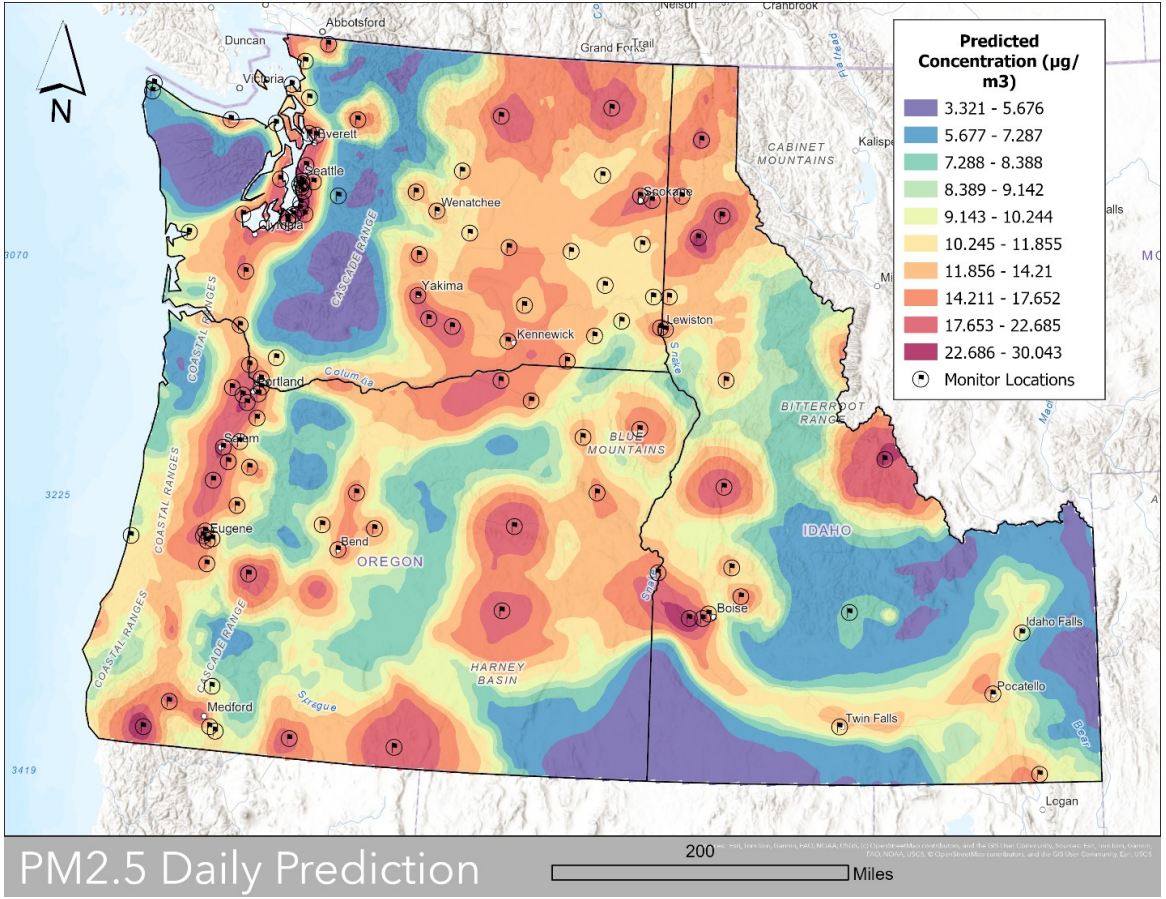


PM2.5 Annual Prediction

Results: Prediction Grids

$PM_{2.5}$ daily

Output range	Monitor input	Model input
3-30 µg/m³	1-42 µg/m³	3-100 µg/m³

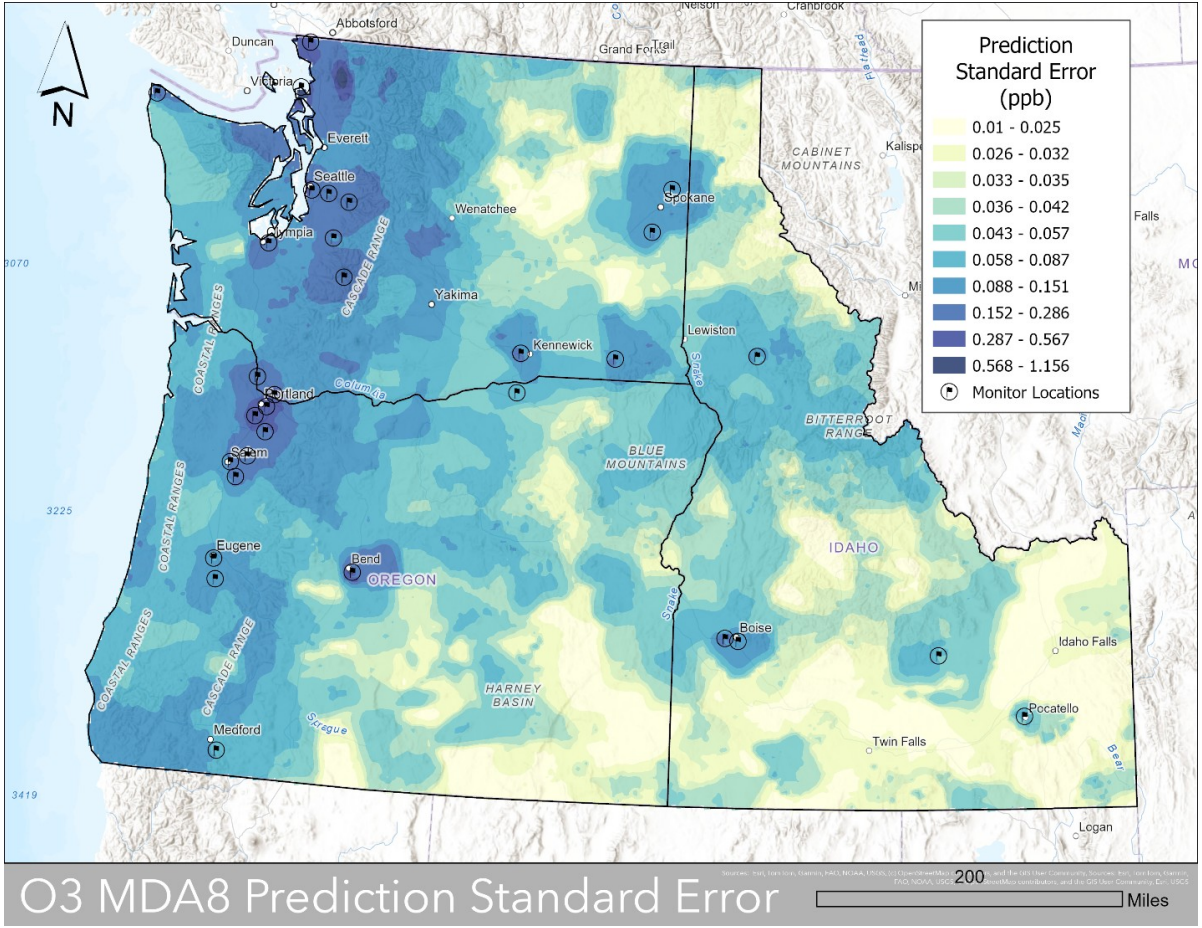


Validation: Standard Error Grids

- Leave-one-out cross validation (LOOCV) assesses the ability of the model to predict values in unknown locations.
- LOOCV is calculated by removing one value from the dataset, re-running the interpolation, and then comparing the predicted value from each run both with and without the value.
- Cross validation statistics in EBK are calculated for each point. Standard error fields are created for the same domain as the predicted surfaces.
- Since the input data is mostly regularly spaced (from the NOAA model grid) and covers the input domain completely, the highest prediction standard errors should correspond to the areas with the highest concentrations.

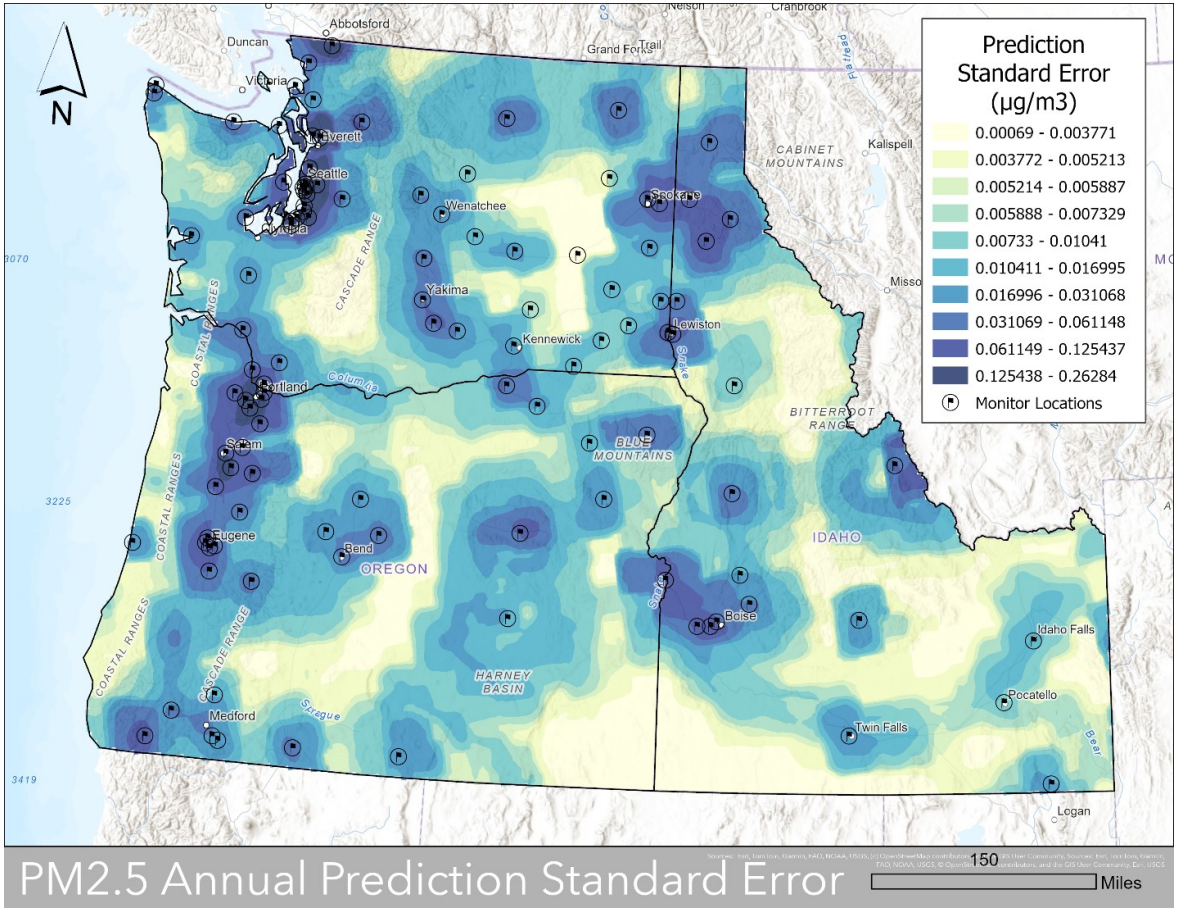
Results: Standard Error

- O_3
- SE range: 0.01-1.156 ppb
- Larger SE correspond with the areas of higher concentration in the prediction map
- SE are very low overall, indicating that the EBK model has a good ability to predict values in unmeasured areas



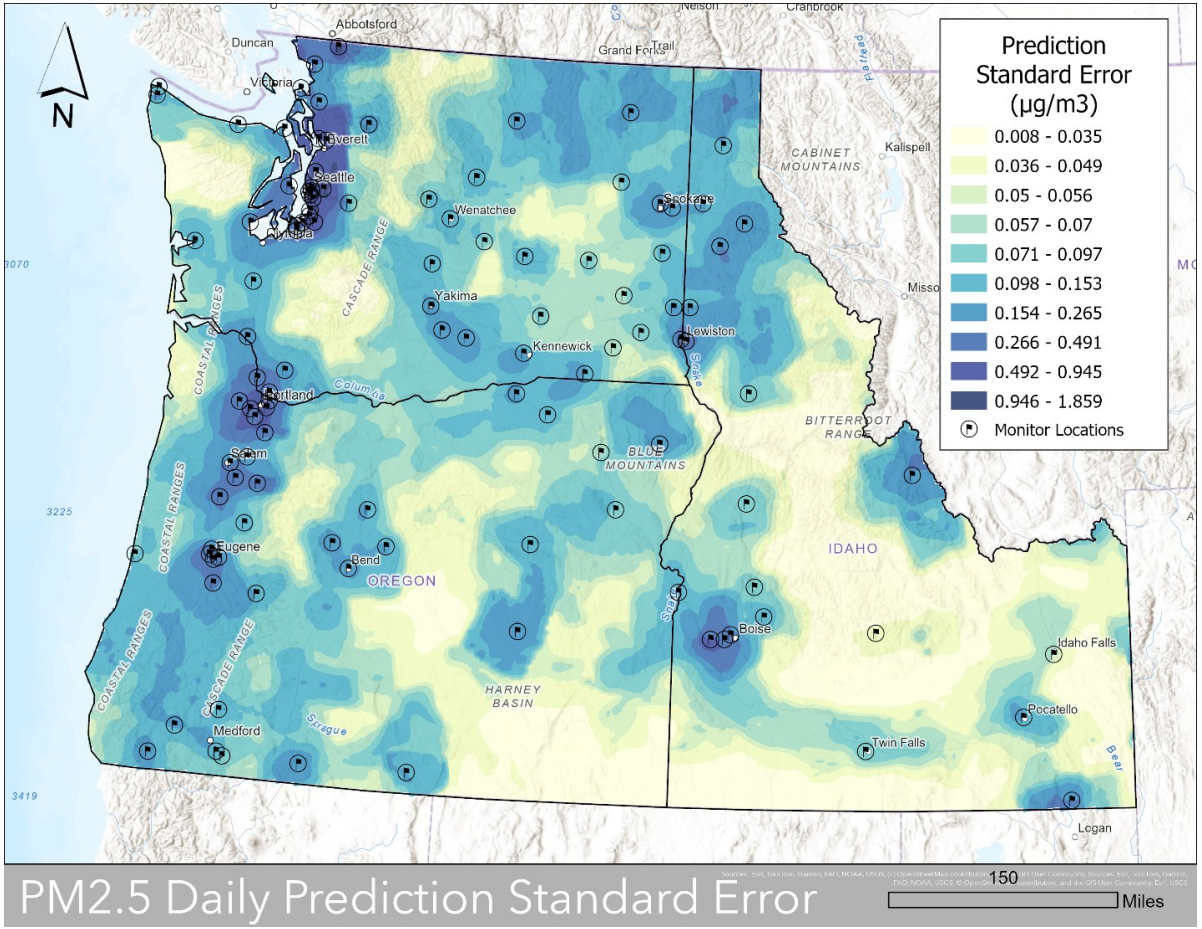
Results: Standard Error

- $PM_{2.5}$ annual
- SE range: 0.001-0.263 $\mu g/m^3$
- Larger SE correspond with the areas of higher concentration in the prediction map
- SE are very low, less than 1 $\mu g/m^3$, which provides high confidence that the EBK model has a good ability to predict values in unmeasured areas



Results: Standard Error

- $PM_{2.5}$ daily
- SE range: 0.008-1.859 µg/m^3
- Larger SE correspond with the areas of higher concentration in the prediction map
- SE are all below 2 µg/m^3

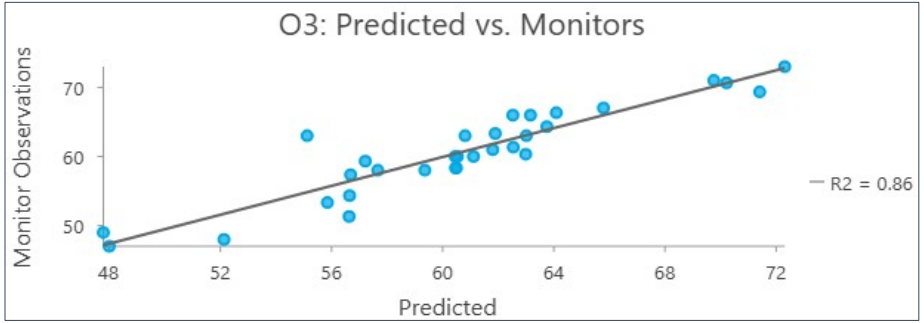


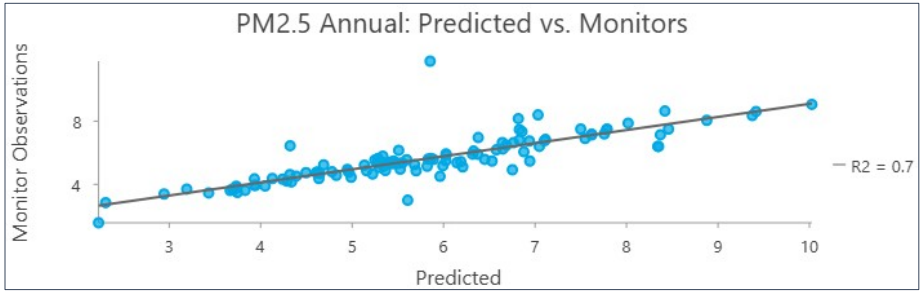
Validation: Scatterplots

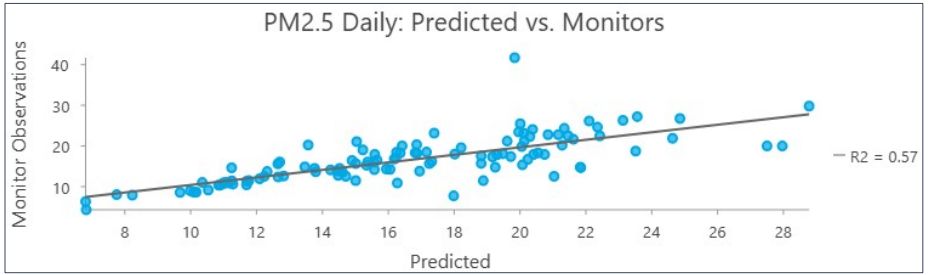
- Comparing the predicted values at monitor sites with the monitor design values
- Comparison reveals:
  - how well the interpolation matches the observations,
  - whether monitor sites are under-or over-predicted,
  - how well the model inputs match the inputs at monitor sites
- Scatterplot outliers
  - If a standalone monitor site has a very different input concentration than the model inputs surrounding it, the interpolation will pull the lone monitor value towards the local mean, thereby smoothing it.
  - Outliers can also occur in areas with high local variability that the interpolation was unable to model accurately.

Validation: Scatterplots

- O3 outliers
  - Bend: 8 ppb underpredicted
- PM2.5 annual outliers
  - Maple Falls, WA
  - Lyons, OR
  - Bellevue, WA
- PM2.5 daily outliers
  - Maple Falls
  - Bellevue
  - Bremerton, OR
  - Seattle area







Discussion

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